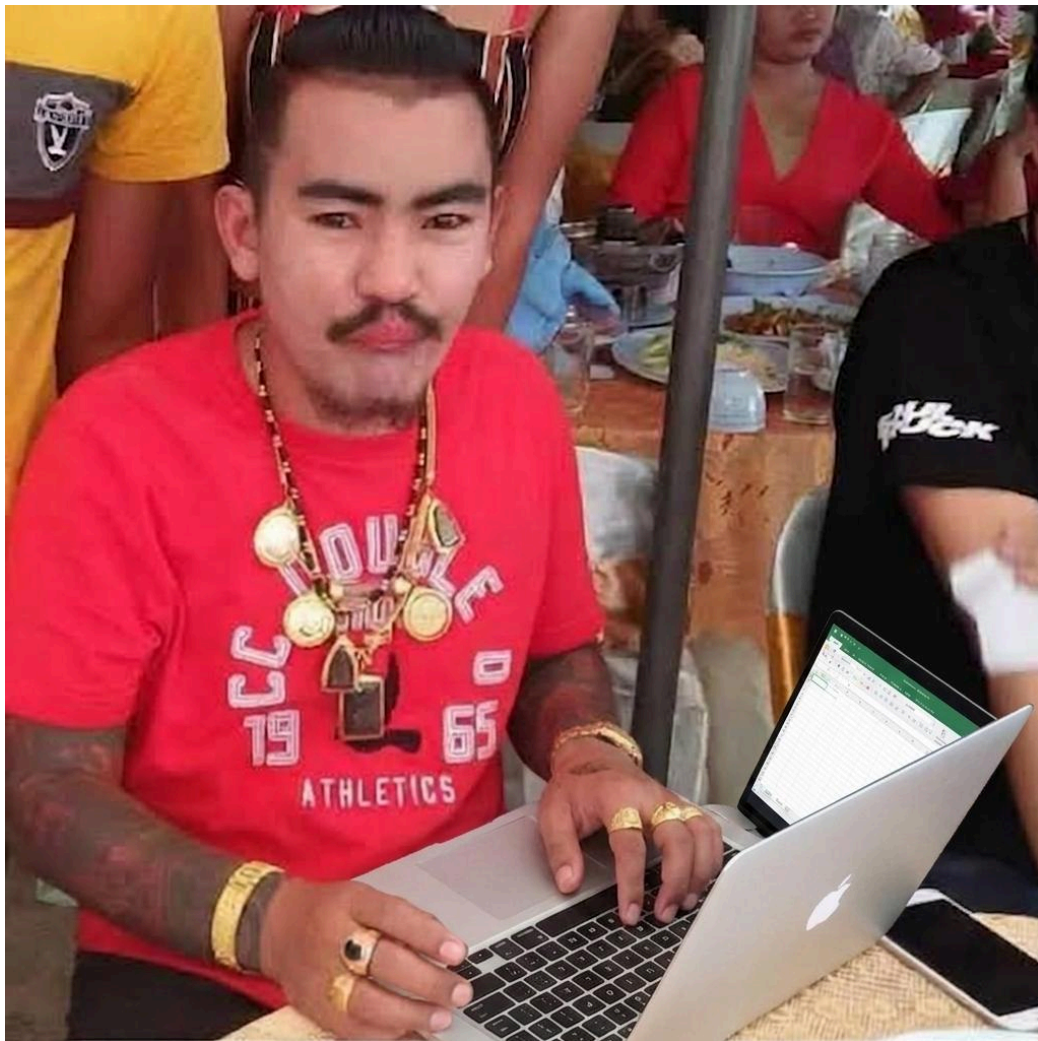


Course: Individual Software Development Process' 2026

Software Requirement Specification

By **ISANPOWER888 (ISP888)**

Chosen Irl Challenge: Project B: Lab Workflow & Request Management



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1. Target Users

- **Lab Member** : Submit requests (e.g., request equipment/space, order consumables, report issues, request access permissions), track the status of their own requests.
- **Teaching Assistant** : Receive and process incoming requests, update status, assign or route requests to the right person.
- **Lecturer / Lab Manager** : Approve requests that require authorisation (e.g., high-cost equipment, special access), view an overview and prioritise requests across the lab, view resource-usage reports.
- **System Administrator**: Manage user accounts and access permissions, maintain system configuration (request categories, statuses, etc.).
- **External Visitor / Industry Collaborator**: A non-member of the lab who requests access to the lab, categorised as either (a) **Open House / General Visit** submitted directly via a public request form without an account, or (b) **Industry Collaboration** submitted on their behalf by a TA/Lab Manager after external coordination (email/phone), since these requests are more formal and often require approval. Both types can track status via a tracking code/link without needing a full account.

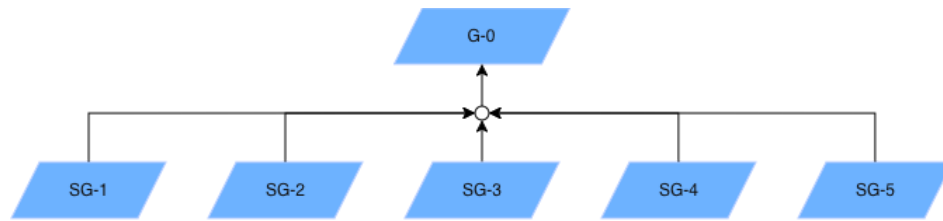
2. The System Goal

G-0: Make lab workflows and requests visible, trackable, and reliable.

The vase lab supports multiple activities (student projects, research, industry collaborations, outreach events). Day-to-day workflows — experiment setup requests, equipment maintenance, access permissions, consumables ordering, and internal tickets (e.g., "fix this script / calibrate sensor") — are currently handled via email threads, chat messages, and ad-hoc spreadsheets. As a result, important requests get lost, priorities are unclear, follow-up is hard to track, and new lab members have no central place to see what's going on or how to request support.

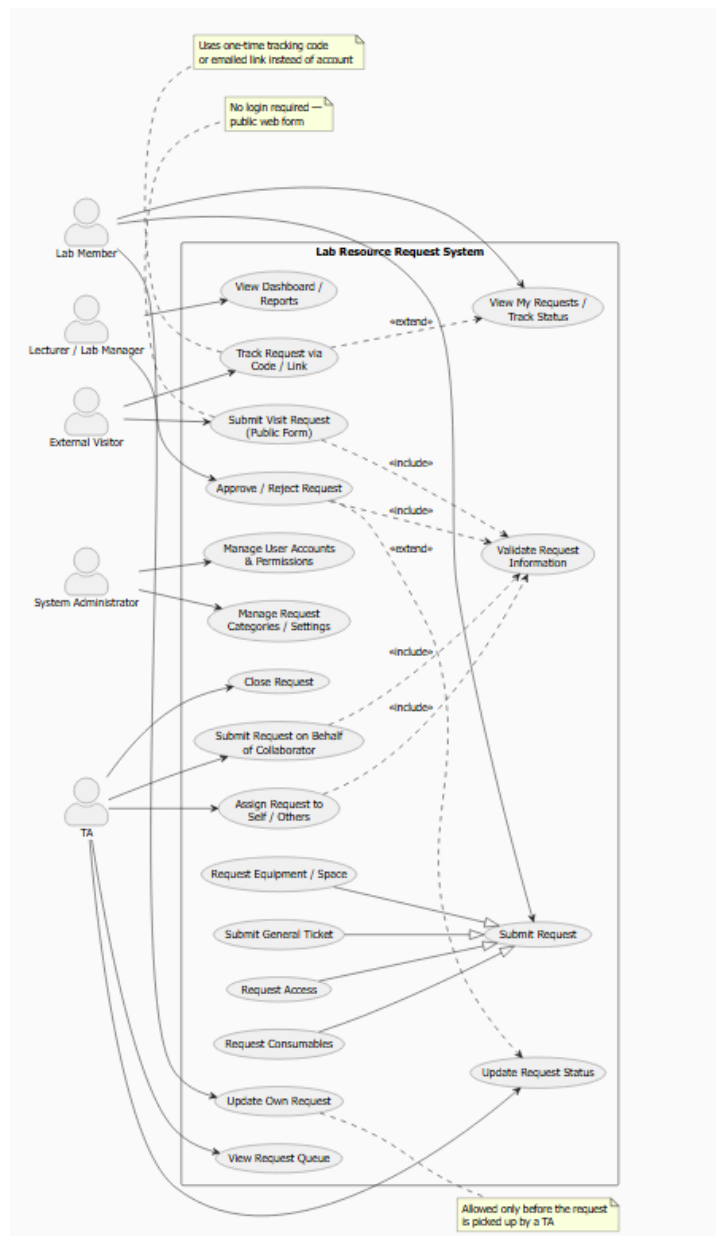
For this iteration, the system scope is limited to **request submission, status tracking, and request handling by authorised staff** for the main request types (equipment/space, consumables, access, general tickets). External/visitor requests (Open House and Industry Collaboration) are included within the "access" request type, with submission via two channels: a public self-service form (no login required) or staff-entry on behalf of the requester. Calendar/booking integration, automated notifications via Line/email, and in-depth analytics reporting are explicitly out of scope for this iteration and will be addressed in later iterations.

3. Goal Refinement via KAOS:



- **Sub-Goal 1 (SG-1)**
Description: Achieve: Allow lab members to submit and categorise requests through a central system.
Rationale: Requests are currently scattered across email, chat, and spreadsheets, causing them to be lost or hard to track.
- **Sub-Goal 2 (SG-2)**
Description: Maintain: Complete and structured information for every request (type, requester, priority, status, timestamps).
Rationale: Complete information allows staff to understand and act on requests correctly.
- **Sub-Goal 3 (SG-3)**
Description: Achieve: Allow authorised staff (TA/Lab Manager) to review, assign, and update the status of requests.
Rationale: Clear ownership is needed to ensure requests are handled and closed.
- **Sub-Goal 4 (SG-4)**
Description: Avoid: Requests being lost or left unaddressed without visibility.
Rationale: This is the core problem stated in the Problem Context important requests currently get lost due to the lack of a central place to track them.
- **Sub-Goal 5 (SG-5)**
Description: Maintain: A traceable history of requests and status changes.
Rationale: Supports auditing, resource-usage analysis, and gives new lab members visibility into past work.
- **Sub-Goal 6 (SG-6)**
Description: Achieve: Allow external visitors and industry collaborators to submit lab-access requests without requiring a full system account, while allowing staff to submit formal collaboration requests on their behalf.
Rationale: The lab serves external stakeholders (open house visitors, industry partners) who fall outside the internal user base but still need a trackable channel into the same request system.

4. Use Case Diagram



5. Use Stories

User Story ID	Description	Acceptance Criteria
US-1	As a lab member, I want to submit a request with a category and description so that TAs/lecturers can see and act on it.	Required fields (type, description, priority) must be filled. A valid request is saved with status "Pending" and appears in the queue.
US-2	As a lab member, I want to view the status of my submitted requests so	The requester can see a list of their own requests with current status and

	that I know if action is being taken.	last-updated time.
US-3	As a TA, I want to view all pending requests in one place so that nothing gets lost or forgotten.	The queue shows all non-closed requests, sortable/filterable by type, priority, and status.
US-4	As a TA, I want to assign a request to myself or update its status so that progress is tracked.	Only authorised users (TA/Lab Manager) can change status. The requester sees the updated status.
US-5	As a lab manager, I want to approve or reject requests that need authorisation so that resource use stays controlled.	Requests marked "needs approval" cannot proceed until an authorised approver acts on them.
US-6	As a lab manager, I want to see a history/report of past requests so that I can understand resource usage over time.	Closed requests remain viewable with full history (who, what, when, status changes).
US-7	As an external visitor, I want to submit an open-house/lab-visit request through a public form without creating an account, so that I can request access easily.	Public form requires name, contact, organisation (optional), visit purpose, preferred date. On submit, systems generate a unique tracking code and show it to the requester.
US-8	As a TA, I want to submit a formal collaboration request on behalf of an industry partner, so that official partnerships are properly logged and routed for approval.	TA can create a request marked "External Industry Collaboration" with collaborator details; request is automatically flagged "needs approval" for Lecturer/Lab Manager.

6. User Requirement Specification

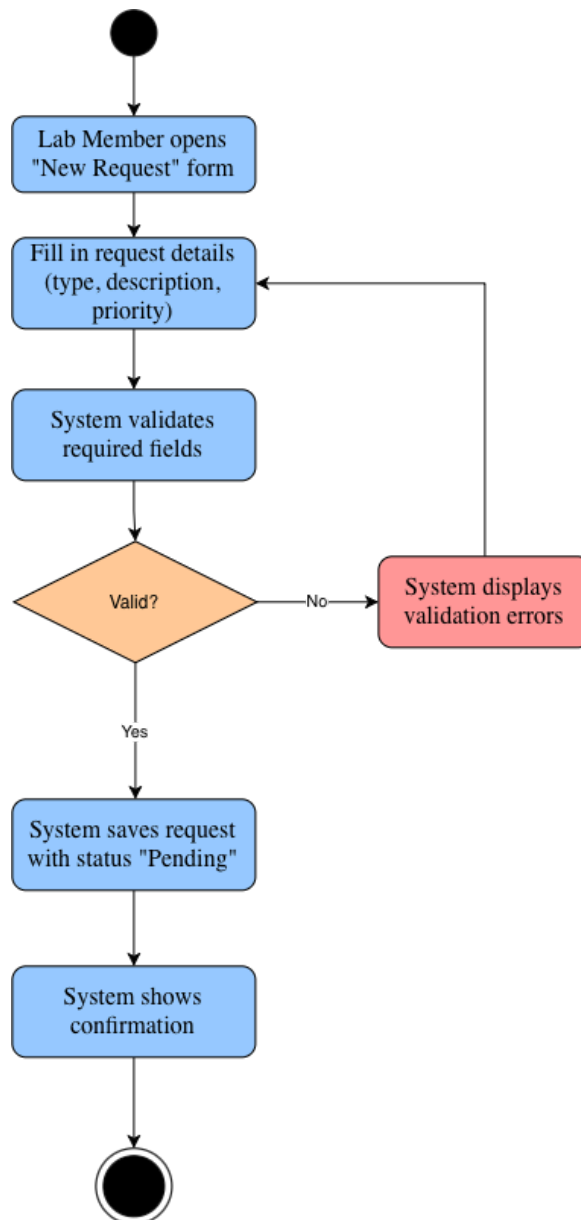
ID	User Requirement
URS-1	Lab members shall be able to submit a request specifying type, description, and priority.
URS-2	Lab members shall be able to view the status and history of their own submitted requests.
URS-3	TAs shall be able to view, filter, and sort all open requests in a central queue.
URS-4	Authorised staff (TA/Lab Manager) shall be able to assign, update the status of, and close requests.
URS-5	The system shall allow designated requests to require approval from the Lab Manager before proceeding.
URS-6	The system shall maintain a traceable record of each request's status changes and timestamps.

URS-7	The System Administrator shall be able to manage user accounts and access permissions.
URS-8	The system shall allow external visitors to submit a lab-visit request via a public form without authentication.
URS-9	The system shall allow TAs to create a request on behalf of an external industry collaborator.
URS-10	The system shall provide external requesters with a tracking code or link to check their request status without logging in.

7. Activity Diagram

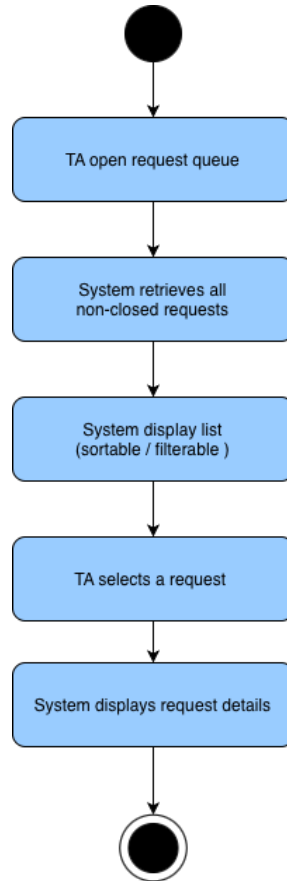
AD-1: Lab Member Submits a Request

This diagram captures the request-submission flow, including field validation, used to derive SRS-3, SRS-4, and SRS-5.



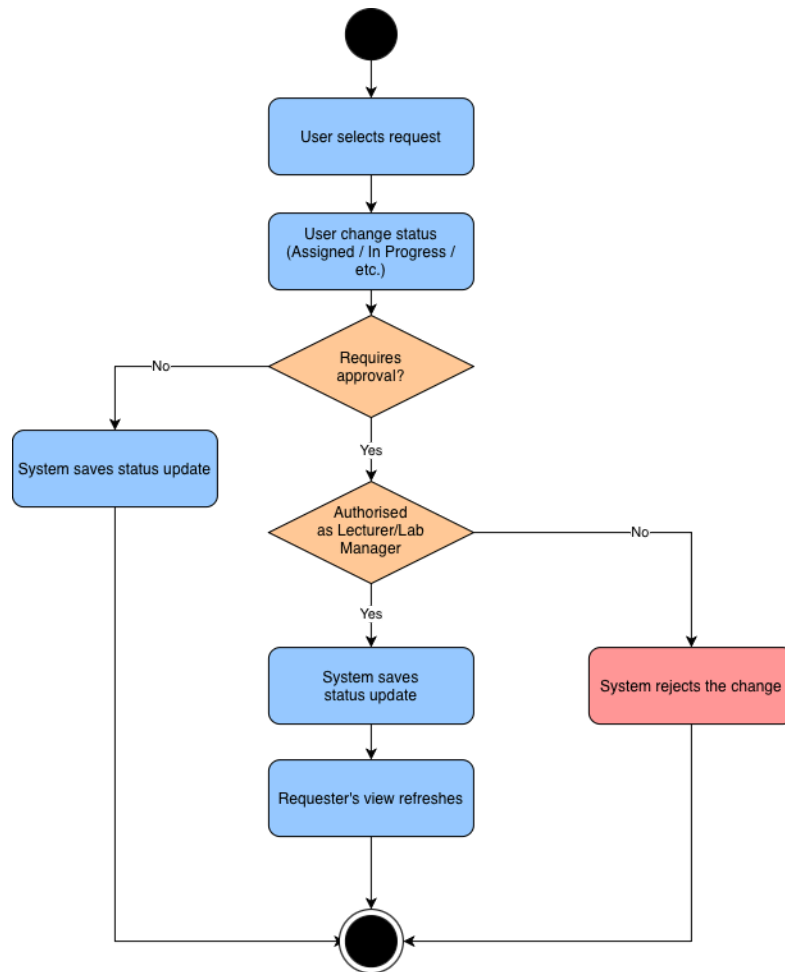
AD-2: TA Views Request Queue

This diagram traces the request-browsing flow used by TAs to derive SRS-1 and SRS-2.



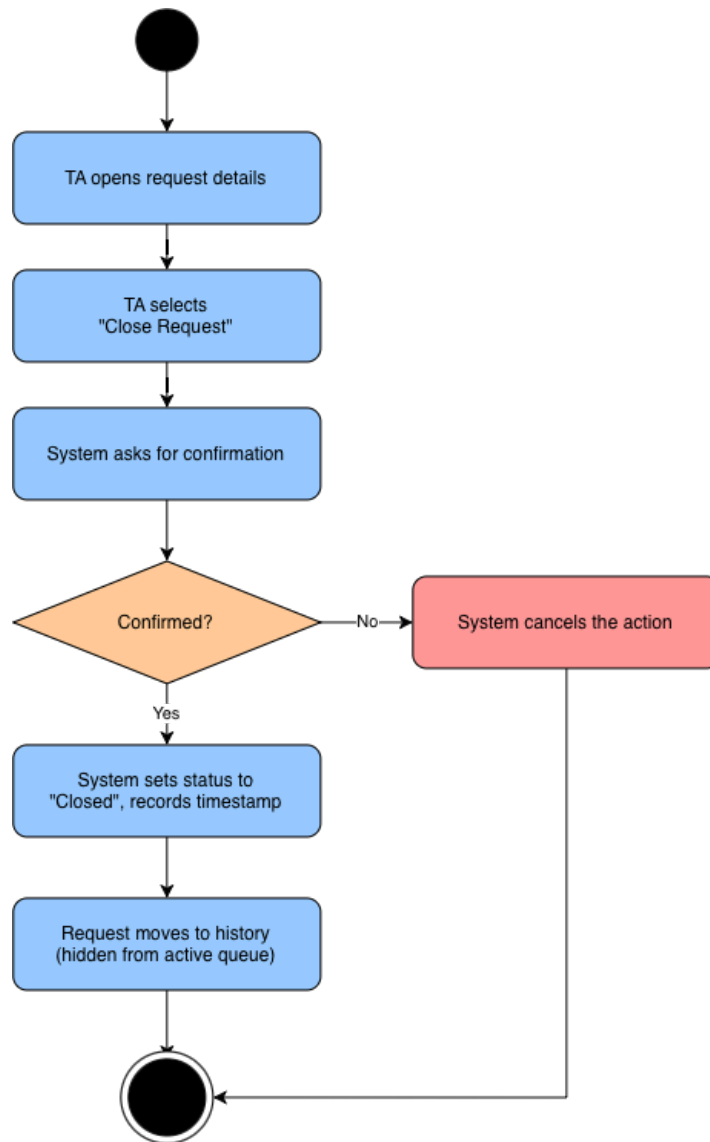
AD-3: Authorised Staff Updates Request Status (with approval check)

This diagram captures the authorisation check and status-update flow (including the approval path) used to derive SRS-6 and SRS-7.



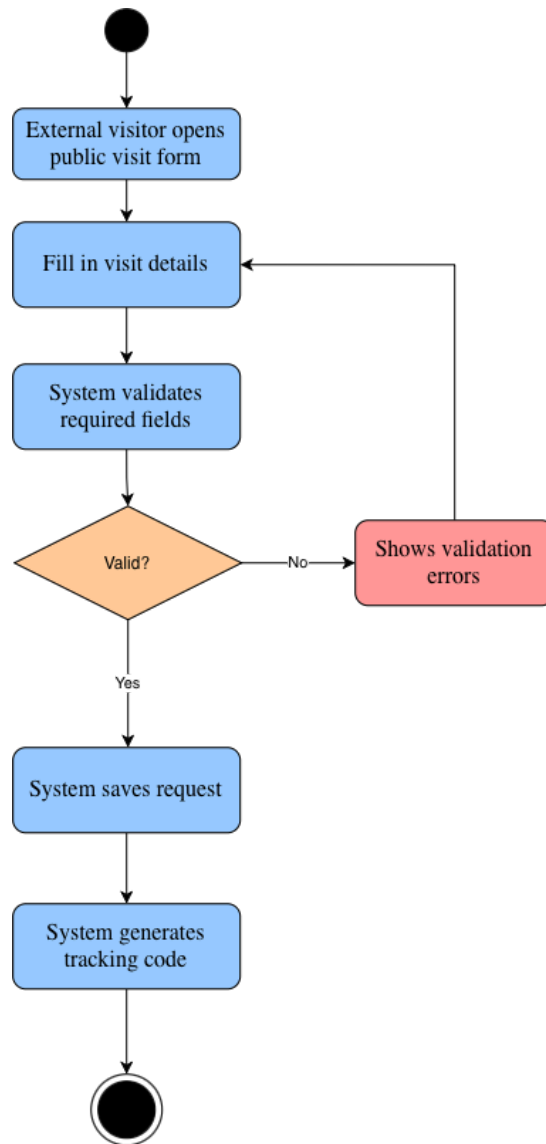
AD-4: TA Closes a Request

This diagram captures the confirmation and archive/close flow used to derive SRS-8.



AD-5: External Visitor Submits Open House Request

This diagram captures the public, unauthenticated request-submission flow used by external visitors (open house / general public), including field validation and tracking-code generation, used to derive SRS-9, SRS-10, and SRS-5.



8. System Requirement Specification

ID	System Requirement
SRS-1	The system shall display only non-closed requests in the active request queue.
SRS-2	The system shall display the type, requester, priority, and status of each request in the queue and in the detail view.
SRS-3	The system shall allow only authorised staff (TA/Lab Manager) to assign, update the status of, or close a request.
SRS-4	The system shall require a request type, description, and priority

	before saving a new request.
SRS-5	The system shall display a validation message and shall not save the request when required data is missing or invalid.
SRS-6	The system shall store each saved request with a status of Pending, Assigned, In Progress, Approved, Rejected, or Closed.
SRS-7	The system shall require Lecturer/Lab Manager approval before a request flagged "needs approval" can move past the Pending status, and shall refresh the requester's view after any saved status change.
SRS-8	The system shall remove closed requests from the active queue while retaining them in a searchable history for reporting.
SRS-9	The system shall allow submission of a lab-visit request via a public, unauthenticated form, capturing name, contact, organisation (optional), purpose, and preferred date.
SRS-10	The system shall generate a unique tracking code for each externally submitted request and display it to the requester upon submission.
SRS-11	The system shall allow an external requester to view their request status by entering their tracking code, without requiring login.
SRS-12	The system shall allow a TA to create a request record on behalf of an external industry collaborator, tagged as "External — Industry Collaboration" and automatically flagged as requiring Lecturer/Lab Manager approval.

9. Traceability Matrix

Sub-Goal	Use Case	User Story	URS	Activity Diagram	SRS
SG-1	UC-1 Submit Request	US-1	URS-1	AD-1	SRS-3, SRS-4
SG-2	UC-1 Submit Request, UC-2 View Request Queue	US-1, US-3	URS-1, URS-3	AD-1, AD-2	SRS-2, SRS-4, SRS-5
SG-3	UC-3 Assign Request, UC-4 Update Request Status	US-4	URS-4	AD-3	SRS-3, SRS-6, SRS-7
SG-4	UC-2 View Request Queue,	US-2, US-3	URS-2, URS-3	AD-2	SRS-1, SRS-2

	UC-5 View My Requests				
SG-5	UC-6 Close Request, UC-7 View Dashboard/Reports	US-6	URS-6	AD-4	SRS-8
SG-6	UC-8 Submit Visit Request (Public), UC-9 Submit on Behalf of Collaborator	US-7, US-8	URS-8, URS-9, URS-10	AD-5	SRS-9, SRS-10, SRS-11, SRS-12